Glossary and Decision Log

UTeM Micro-Credential Management System (MCMS)

Document Control

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Glossary

Term	Definition
ABC	Academic Bank of Credit. Central digital depository of ACR. A deposit is storage and does not approve transfer. [G 12.5]
ACR	Academic Credit Record. Academic record used in the ABC mechanism; prepared only for an eligible completed UMC or Quality-Verified SAMC. [G 12.4]
Achievement Record	UTeM internal record of modules passed, Digital Badges and MC completion (Rekod Pencapaian Pembelajaran). It is not an ACR. [G 8.7.5, 18.3]
Adapter	A defined interface to an external system (UCreds, ABC, QVC/MQA, payment, SSO, email) with a stub or manual version in the first release.
API	Application Programming Interface
Audit trail	Append-only record of approvals, credential actions, result changes and corrections. History is never erased. [G 10.5.1, 18.10]
CODL	Centre for Open and Distance Learning. Assumed owner of MCMS and operator of UCreds.
Configurable value	A threshold or list kept in settings and never hard-coded (40 h SLT, 3-year QV limit, 10-year credit validity, 30% home credit, residency, alert days, retention).
CQI	Continuous Quality Improvement: finding, evidence, analysis, action, implementation and effectiveness check. [G 17]
Credit accumulation	Building a record of eligible credit over time. Separate from credit transfer. [G 12]
Credit transfer	Receiving programme's evaluation and acceptance of credit. Equivalence is judged on learning outcomes, not on credit or SLT alone. [G 13]
Credit validity	Credit in ABC is valid 10 years from deposit date. This differs from ACR retention (indefinite unless the owner or authority directs otherwise). [G 12.8]
Digital Badge	Proof of achievement of one module or unit. Not academic credit and not an ACR. [G 8.7.1, App E1]
Digital Credential / Certificate	Proof of completing the whole MC (Sijil / Digital Credential). Does not by itself give credit or transfer. [G 8.7.3, App E1]
HEPA	Student Affairs and Alumni Department. Assumed to register outside applicants together with Pejabat Pendaftar.
Material change	Change to outcomes, SLT, assessment, module structure, credit, MQF level, parent course link or QV status. Needs review and re-approval and is versioned. [G 9.5, 17.7, 17.8]
MCE	Micro-Credential Eksekutif. Non-credit short learning under 40 hours SLT. Never carries credit unless reclassified and re-approved as a SAMC. [G 6.3.3]
MQA / MQF	Malaysian Qualifications Agency / Malaysian Qualifications Framework.
Micro-Credential (MC)	Certification of a small, focused unit of learning with defined outcomes and assessment. Recognition follows achievement, not attendance. [G 6.1]
Module	A smaller unit of a UMC. Completion earns a Digital Badge, not credit and not whole-UMC completion. [G 8.7.1]
PTj	Pusat Tanggungjawab: the faculty, school, institute or centre that owns an offering. [G 15.2.2]

Term	Definition
QA / QV / QVC	Quality Assurance applies to every MC. Quality Verification is the SAMC-specific check run by the QVC under MQA rules. Approval to offer is a separate decision from QV. [G 7.6, 11.2]
Quality-Verified SAMC	A SAMC that has passed QV. QV is valid for at most 3 continuous years and needs one completed cohort before submission. [G 11.2.1, 11.2.2]
RBAC	Role-Based Access Control
SAMC	Stand-alone Micro-Credential. Own offering, at least 40 hours SLT. Credit and MQF level exist only after QV. [G 6.3.2]
SDD	Software Design Description
Single Redemption	Credit used for one qualification cannot be used for another. Transfer makes credit provisionally redeemed; completing the programme makes it redeemed. [G 12.9]
SLT	Student Learning Time: all learning and assessment time, not only contact time. [G 8.3]
Stackable Degree	Planned pathway where eligible credit counts toward part of a degree. Needs admission, transfer approval, at least 30% credit earned at UTeM, residency (1 semester for conventional programmes), PLO mapping and graduation requirements. [G 14]
SRS / PRD	Software Requirements Specification / Product Requirements Document
SSO	Single Sign-On
UCreds	UTeM online learning platform for micro-credentials; the pilot learning platform. Reached only through an adapter.
UMC	Unbundled Micro-Credential. Made by unbundling an accredited credit course; keeps its parent link; total SLT equals the parent course; whole UMC is at least 40 hours SLT or one credit. [G 6.3.1]

1. Introduction

1.1 Purpose

This document fixes the vocabulary and the key decisions that the three controlling documents of the UTeM Micro-Credential Management System (MCMS), namely the Software Requirements Specification (SRS), the Product Requirements Document (PRD) and the Software Design Description (SDD), shall follow. Every later document shall use the terms in the Glossary with exactly these meanings and shall not contradict a decision in Section 3 unless the decision is first changed here and the version of this document is raised.

The document set is written for two readers at once: a supervisor who reviews it, and AI development agents and developers who use it as the guide, reference and rule set for building the system. For that reason requirements are written as single, identified statements with traceability to the Guide.

1.2 Document Conventions

This document and the document set follow these conventions:

Term	Description
SHALL	Mandatory requirement that must be fulfilled in Release 1, the first buildable release (prototype scope, decision D-22). The developer is required to cover this feature in Release 1.
SHOULD	Requirement planned for Release 2. The developer should take it into account in the design so that it can be added without rework.
MAY	Requirement anticipated for Release 3 or later. The developer is encouraged to keep it in mind for future scalability.
TBD	To Be Determined. Information not yet available that will be provided in a later version. Every TBD is linked to an open item (OI-xx).
Note	Provides additional information or clarification.

> **Note:** A requirement tagged with a Guide source, for example [G 12.9], states a rule that is mandatory for the complete system under UTeM policy. Its SHALL / SHOULD / MAY wording states only the release in which MCMS implements it. Guide rules are never relaxed by the release plan.

Requirements and records are categorised as follows:

Requirement Number	Description
FR-XX	Functional Requirements
NFR-XX	Non-Functional Requirements
IR-XX	Interface Requirements
DR-XX	Data Requirements
SR-XX	Security Requirements
BR-XX	Business Rules taken from the Guide (added to the reference categories, decision D-25)
D-XX / A-XX / OI-XX	Decision / Assumption / Open item (this document)

Source tags follow the Guide: [G 12.9] means Guide clause 12.9 and [App E5] means Guide appendix E, table E5.

1.3 Intended Audience

This document is intended for the following stakeholders:

Stakeholder	Role
Supervisor	Reviews the document set and the decisions taken.
CODL / UTeM reviewers	Confirm assumptions and open items; system owner (assumed).
Development team and AI development agents	Design and implement the system using these terms and decisions as fixed rules.
Testers and quality assurance	Derive acceptance tests from requirement IDs and rules.
System administrators	Maintain and configure the system after delivery.

1.4 Scope of this Document

In Scope

- Glossary of terms used across the SRS, PRD and SDD
- Key concepts of the Guide that drive the design (categories, recognition stages, numeric rules)
- Decision log, assumptions register and open items with the working assumption for each
- Change control for this document

Out of Scope

- Requirements, user needs and design detail, which belong to the SRS, PRD and SDD
- A Test Plan (acceptance tests are traced from SRS requirement IDs; decision D-01)
- Any change to the Guide itself

1.5 References

1. Garis Panduan Micro-Credential UTeM (the Guide), CODL UTeM, with Lampiran A to H. Controlled version to be confirmed (OI-01).
2. Software Requirements Specification, UTeM Micro-Credential Management System, informal draft v0.1, 23 October 2026 (input material only).
3. Software Requirements Specification for an AI-Driven Job Matching Platform, v1.0, May 2025. Used as a format reference only; no content is reused.
4. MQA: Guidelines to Good Practices on Micro-Credentials; Quality Verification of Stand-alone Micro-Credentials; Guidelines on the Academic Bank of Credit; Malaysian Qualifications Framework (second edition).
5. Guidelines on Developing Academic Programme through Stackable Degree Pathway.
6. ISO/IEC/IEEE 29148, Requirements engineering.
7. Open Badges 3.0 and W3C Verifiable Credentials data model (guidance for credential format).
8. WCAG 2.1; OWASP Application Security Verification Standard; Personal Data Protection Act 2010 (Act 709) with current amendments.
9. Software Requirements Specification, UTeM Micro-Credential Management System, v1.0 draft, 23 October 2026 (MCMS_SRS_V1.0_Oct_2026), written against v0.2 of this document; source of open items OI-09 to OI-15.

2. Key Concepts from the Guide

This section summarises the parts of the Guide that shape every design decision. It does not replace the Guide.

2.1 Micro-Credential Categories

Criterion	UMC	SAMC	MCE
Origin	Unbundled from an accredited credit course; keeps parent link	Own offering, not part of an accredited programme	Own non-credit short learning
SLT	Total equals parent course; whole UMC at least 40 h or 1 credit	At least 40 h	Under 40 h
Credit	Possible, subject to academic rules; only the whole UMC carries credit	Only after QV	None
Quality route	Quality assurance of parent programme	QA, plus QV if credit is sought	Internal UTeM QA
ABC deposit	Yes, if eligible	Only if Quality-Verified	No
Stackable degree	Yes, subject to conditions	Only if Quality-Verified	No

Source: Guide Table 1 and clauses 6.3 to 6.5.

2.2 The Separate Recognition Stages

The Guide treats each stage below as a different thing with its own decision and its own record [G 12 to 14, 18.14]. MCMS shall store them as separate records so that none can be skipped or mistaken for another.

Stage	Decided by	Does NOT imply
1. Approval to offer	Senate (assumed), after faculty processing	Credit, QV, transfer
2. Module achievement (Digital Badge)	Assessment verified by the owning PTj	Whole-MC completion, credit
3. MC completion (Certificate / Digital Credential)	PTj; credential issued by the records unit	Credit, ACR, transfer
4. Quality Verification (SAMC only)	QVC / MQA	Transfer or admission
5. ACR and ABC deposit	Records unit prepares; ABC depositor deposits; participant consents	Transfer approval
6. Credit transfer	Receiving programme / faculty	Admission, degree award
7. Stackable degree award	UTeM academic programme	Nothing further

2.3 Key Numeric Rules (Configurable)

ID	Rule	Default	Source
BR-01	Minimum SLT for SAMC and for a whole UMC	40 hours (or 1 credit for UMC)	G 6.3.1, 6.3.2
BR-02	Maximum validity of Quality Verification	3 continuous years	G 11.2.2

ID	Rule	Default	Source
BR-03	QV submission needs at least one completed cohort	1 cohort	G 11.2.1
BR-04	Validity of credit in ABC from deposit date	10 years	G 12.8
BR-05	Minimum credit earned at UTeM for a stackable degree	30% of graduating credit	G 14.5.1
BR-06	Residency for conventional programmes via stackable pathway	1 semester	G 14.5.3

3. Decision Log

3.1 Status Key

Decided: stated by the project owner or fixed by the Guide. **Assumed**: no answer yet; a working assumption is used and its effect if wrong is recorded in 3.3. **Proposed**: a recommendation awaiting confirmation. Every Assumed or Proposed item shall be built so that it can change through configuration or by replacing one adapter, not by rewriting modules.

3.2 Decisions

ID	Decision	Rationale / effect if changed	Status
D-01	The controlling document set is the SRS, the PRD and the SDD. A Test Plan is not in this set; acceptance tests are traced from SRS requirement IDs.	Matches the project owner's choice. A Test Plan can be added later without changing the three documents.	Decided
D-02	All documents are written in English. The system (screens, notifications, credentials) shall support Bahasa Melayu and English.	Documents are read by a supervisor and by AI agents. Bilingual wording of the Guide is preserved through the Glossary.	Decided
D-03	The Guide is treated as Version 2.0 Final, 15 September 2026 (file name) until CODL confirms the controlled version. Its cover page says Version 1.0, 18 September 2026.	Clause numbers may shift between versions. If the other version controls, only source tags need re-checking.	Assumed (OI-01)
D-04	A new, formal SRS shall be written. Its layout follows the reference SRS (document control, contents, glossary, numbered sections, identified requirement tables) with Guide terminology. The informal draft SRS v0.1 is input material only.	The draft ends at section 2.8 and its later sections do not exist. A new SRS avoids inheriting the gaps.	Decided
D-05	All documents are formal and professional in style and written so an AI agent can follow them without guessing: stable IDs, one requirement per statement, traceability to Guide clauses, explicit assumptions.	Dual audience: supervisor review and AI-assisted development.	Decided
D-06	MCMS is a new standalone system with its own database, login handling and workflow. It talks to other systems only through adapters.	Carried from draft SRS (C1, C3). Keeps UCreds and the student system untouched.	Decided
D-07	Architecture is a modular monolith: one deployable application with clearly separated modules (starting from the draft's M1 to M16, regrouped in the SDD if needed).	Simplest to build, test and host for a small team. Modules use internal interfaces, so later splitting stays possible.	Proposed
D-08	Primary technology is Laravel (PHP).	CODL mostly uses PHP, so maintenance stays with a familiar stack. Chosen by the project owner.	Decided
D-09	Database recommendation: PostgreSQL as the primary choice; MySQL or MariaDB is an acceptable fallback if CODL already hosts and supports it.	PostgreSQL gives strong constraints, transactional integrity, JSON fields for credential metadata and good support for append-only audit design. The SDD keeps queries portable through Laravel's database layer.	Proposed (OI-02)

ID	Decision	Rationale / effect if changed	Status
D-10	Identity: staff and participants sign in with their Microsoft education account through single sign-on (OpenID Connect assumed). Outside applicants register first at Pejabat Pendaftar or HEPA and then receive that account. MCMS offers no open public sign-up.	Stated by the project owner with details uncertain. If the method differs, only the SSO adapter and applicant recognition change.	Assumed (A-06, OI-03)
D-11	Approvals: the Senate approves offerings, credit transfer rules and stackable mapping; faculty/PTj processes proposals, verifies category and decides individual transfer cases. Approving bodies are settings, not code.	The Guide deliberately names no committees (App H). If bodies differ, workflow routing changes by configuration.	Assumed (A-01)
D-12	The records and credential unit (P6) and the product owner are CODL. Role holders are set in configuration.	Placeholder for the owner's uncertainty. Reassigning a role must not need a code change.	Assumed (A-07)
D-13	UCreds is the pilot learning platform. Its connection method is a placeholder for the next iteration; CSV import is the fallback. UCreds is assumed to issue module badges and keep its own database; MCMS imports and checks badge data and issues certificates and Digital Credentials.	If UCreds does not issue badges, MCMS issues them instead (one adapter and one module change).	Assumed (A-04, A-05, OI-04)
D-14	ABC, QVC/MQA submission and payment are manual or stubbed in Release 1: MCMS prepares and tracks; UTeM staff submit. A dummy payment service stands in.	These external services are not ready or not integrable now. Each is replaced later by swapping one adapter.	Assumed (A-02, A-03)
D-15	Recognition stages (Section 2.2) shall be stored as separate records with separate decisions. No single status or button may combine them.	The single most important rule in the Guide [G 12 to 14, 18.14]. Overrides any convenience shortcut in later documents.	Decided
D-16	Thresholds in Section 2.3 and other limits shall be configurable settings, including alert days and retention periods.	Rules in the Guide may be revised by UTeM or MQA. Carried from draft SRS (C4).	Decided
D-17	Credit value, MQF level and QV wording shall be generated only from verified records, never typed by hand. A credential shall show credit, MQF or QV information only when verified and still valid.	Prevents false claims [G 10.5, App E5].	Decided
D-18	The draft's rule C9 ("a UMC module is never a credit unit") is corrected: a UMC module is not a credit unit unless specifically approved as a separately recognisable credit unit.	Guide 13.5 and 14.2.1 allow this exception. The system shall support an approval flag whose default is no.	Proposed (OI-05)
D-19	Credentials use an open, signed format (Open Badges 3.0 and W3C Verifiable Credentials as guidance); blockchain is not required. Public verification needs no login and shows only the minimum permitted information.	Carried from draft SRS (C8). Privacy limits follow Guide App E6.	Proposed
D-20	All approvals, credential actions, result changes and corrections are written to an append-only audit log. Corrections never delete earlier records.	Required by G 10.5.1, 15.7 and 18.10. Also supports supervisor review.	Decided
D-21	Sizing figures (up to 200 offerings and 20,000 participants a year, peaks of 2,000 registering within hours) are indicative only and non-binding.	The project owner considers them a possibility, not fact. Design for modest scale without blocking growth.	Assumed (A-09)

ID	Decision	Rationale / effect if changed	Status
D-22	The system is produced by an intern as a mock-up and concept. The documents describe the target system. Release 1 is the first buildable release (prototype scope); the PRD shall mark which features belong to Release 1, 2 and 3 or later.	Keeps the specification complete while being honest about prototype scope. Release numbers drive **SHALL** / **SHOULD** / **MAY** (D-24).	Decided
D-23	Deliverables are produced as PDF, one at a time, in this order: this document, SRS, PRD, SDD. Each is reviewed before the next begins.	Lets the supervisor check decisions early and lets each later document build on confirmed text.	Decided
D-24	**SHALL**, **SHOULD** and **MAY** indicate Release 1, Release 2 and Release 3 or later respectively, as in the reference SRS. Guide-mandated rules keep their [G] tag and stay binding on the full system regardless of release.	Follows the reference format while preventing the release plan from weakening any Guide rule. Replaces the plain shall / should / may meaning used in v0.1.	Proposed
D-25	A sixth requirement category, BR-XX (Business Rules from the Guide), is added to the reference categories FR, NFR, IR, DR and SR.	The Guide's rules are central to this system and need their own traceable IDs.	Proposed
D-26	Format of all documents follows the reference SRS: cover with Document Control table and disclaimer, table of contents, glossary, numbered sections with tinted headings, serif body text, and tables with ID and Requirement columns. The PRD and SDD adopt these elements where applicable and use their own section content.	Requested by the project owner. Only structure and layout are adopted; no content from the reference is reused.	Decided

3.3 Assumptions Register

ID	Assumption	Effect if wrong
A-01	The Senate approves offerings, transfer rules and stackable mapping; faculty process each case.	Approver bodies are settings; workflow routing changes without a rebuild.
A-02	ABC and QVC/MQA accept submissions manually in Release 1.	A real adapter is added later.
A-03	UTeM's payment hub is not ready; a dummy payment service is used.	Replace one adapter.
A-04	UCreds is the pilot learning platform; its connection method is unknown.	CSV import is the fallback; connector design shifts.
A-05	UCreds already issues module badges and has its own database.	MCMS issues badges itself if UCreds should not.
A-06	Outside applicants register at Pejabat Pendaftar or HEPA, then sign in with a Microsoft education account.	Add an applicant pre-registration screen.
A-07	The records and credential unit (P6) and the product owner are CODL.	Role holders change in settings.
A-08	Retention periods are placeholders.	Edit settings.
A-09	Sizing is indicative: 200 offerings and 20,000 participants a year, peaks of 2,000 registering within hours.	Capacity plan changes.

3.4 Open Items for CODL / Supervisor Confirmation

OI-01 to OI-08 were raised in v0.1. OI-09 to OI-15 were found while writing the SRS v1.0 draft and are added in v0.3; the SRS requirements they affect are listed in its Appendix C.

ID	Question	Needed by	Working assumption until answered
OI-01	Which Guide version is controlling: V2.0 (15 Sep 2026) or V1.0 (18 Sep 2026)?	Before SRS approval	V2.0 (D-03)
OI-02	Is a PostgreSQL server available, or must the system use MySQL/MariaDB?	Before SDD	PostgreSQL, portable queries (D-09)
OI-03	Which SSO protocol and tenant settings does UTeM allow for the Microsoft education account? How are outside applicants recognised?	Before SDD	OpenID Connect; account attribute or imported list (D-10)
OI-04	How does UCreds connect (web service, data export, Moodle link) and does it issue badges?	Next iteration	CSV import; UCreds issues badges (D-13)
OI-05	Is a separately credit-bearing UMC module an approved case at UTeM, and who approves it?	Before SRS approval	Supported by flag, default no (D-18)
OI-06	Which committee or role approves each decision (offering, transfer rules, stackable mapping, material change)?	Before SDD	Senate approves, faculty processes (D-11)
OI-07	Retention periods for each record category.	Before go-live	Placeholder settings; ACR retained indefinitely [G 12.8]
OI-08	Who owns the credential signing keys and the public verification page?	Before SDD	CODL (D-12)
OI-09	What are the credit transfer limits and non-transferable components of each target programme, and who maintains them? [G 13.5]	Before Release 1 testing	Programme settings entered by the faculty; the system warns only
OI-10	How will MCMS learn of programme admission, completion and withdrawal, which drive the redemption status of credit? [G 12.9, 14.4]	Before SDD	Recorded manually by an authorised user from programme records
OI-11	What data and file format does ABC require for an ACR deposit, and how are redemption updates sent to ABC? [G 12.5, 18.6]	Before Release 1 testing	Export of the ACR fields of Appendix F3 in CSV and PDF
OI-12	What fee rules, refund rules and sponsorship arrangements apply to offerings?	Before Release 2	Fee set per offering; refunds handled outside MCMS
OI-13	What is the completion window for a modular UMC ("tempoh kesahan pembelajaran", G 7.4, 12.2)?	Before Release 1 testing	Set per offering; no default
OI-14	Confirm that 40 notional learning hours equal 1 credit for UMC and SAMC calculations. [G 6.3.1]	Before SRS approval	40 hours = 1 credit (MQF)
OI-15	Which identity verification methods are required for online assessment at each risk level? [G 10.4]	Before Release 1 testing	Recorded per assessment; method chosen by the PTj

4. Change Control

This document is the single place where terms and decisions change. A change shall state the decision ID, the new wording, the reason and the date, and shall raise the document version. The SRS, PRD and SDD shall cite the version of this document they were written against.

Version	Date	Change
0.1	23 October 2026	First issue: glossary, decisions D-01 to D-23, open items.
0.2	23 October 2026	Restyled to follow the reference document format; added Key Concepts, assumptions register, change history, decisions D-24 to D-26; D-04, D-05 and D-22 reworded.
0.3	23 October 2026	Added open items OI-09 to OI-15 raised while writing the SRS v1.0 draft. "Needed by" of OI-01 and OI-05 changed from "Before SRS" to "Before SRS approval", since the SRS draft now exists and uses their working assumptions. Added the SRS v1.0 draft to the references. The current assumptions (release scope, indicative performance figures) are accepted by the project owner for now. No decision changed.